Project Proposal

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Project Proposal

Context Aware Recommender System for London Based Restaurants

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1.0 Introduction

A rapid growth of information technologies such as websites, applications and social networks brought many benefits such as broad product and service offering for consumers as well as increasing business development opportunities for retailers and service providers. Consequently, a downside and a challenge of the phenomenon has become the information overload, mainly in the context of personalisation and prioritization from a consumer's point of view. [2], [17]

Recommender Systems have become a crucial tool and technique which optimizes and accelerates the process of purchasing goods and services [11] amongst an enormous number of offerings. They tackle the necessity to meet growing consumers' expectations for a smarter, more suggestive, quicker and personalised offering. RS have started playing a critical role in society [24] therefore significant advances in computational intelligence drive research and work related to RS refinement and improvement. This proposal presents a plan to develop a restaurant Recommender System which combines solutions that tackle issues and shortfalls identified so far and focuses on tackling critique expressed by consumers.

2.0 Background

2.1 Overview of Recommender Systems

Having become an increasing area of interest in mid-1990s [11] 'RS systems are concerned with data collection, pre-processing, transmission, data storage and extracting information using statistical and analytical methods.' [16] The fundamental models of RS are built using the following methods: [2], [16]

- Collaborative filtering (CF): based on users' interaction with the items through rating evaluation or purchase behaviour where recommendations are made based on choices and opinions of similar users.
- Content-based filtering (CB): based on user profile and item attribute information where selected keywords and descriptions are matched, and recommendations are made based on items that user liked or purchased in the past.
- Hybrid: based on a combination of several methods deriving from the basic models of RS.

Driven by various factors such as known drawbacks of traditional models, evolution of the web and increased accessibility to additional explicit and implicit data RS introduced new trends, namely context-aware recommender systems (CARS). Traditional RS task has been to predict a user's rating for a chosen item using a 2-dimensional model (user * item) [4]. CARS however are based on a multi-dimensional model. Linking context or supplementary information like contextual data (information time, the location social context) CARS overcome 2-dimesnional model's knowledge-poor technique and presents a richer knowledge-dependent method. [2]

2.2 Recommender System Limitations

Data sparsity: missing information given a new user and/or a new item [25]

Cold-start problem: system's limitations on recommending a product to a new user that has no past information or a new item with no or little rating. [3]

Scalability: an instance where the number of users significantly exceeds the number of items. [25]

Shilling Attacks: large amount of positive ratings and reviews produced by users for their own materials and negative ones for their competitors. [25]

Personal privacy: recent issues with personal data and links to social networks. [10]

Explainability: consumers losing confidence in recommendations engines and businesses influencing the own rankings and ratings. [22]

2.3 Recommender System architecture and evaluation

2.3.1 Choosing the model

Practitioners and researchers have recognised that generally CF and/or combination of CF and other models are the most popular and successful approaches for RS [25], [9] widely adopted in retail and entertainment applications. Travel and hospitality products are considered to be of a more complex nature – they often have elements of intangibility and are associated with experiences and feelings. CF is considered to simulate word-of-mouth recommendation, however, a disadvantage of many current applications of CF-based travel recommender is limitation on providing multi-class recommendations. [2] To overcome these issues, CARS is considered to be the most suitable multi-dimensional model where contextual information can consist of specific location, time of purchase/visit, social status, or weather conditions [2], (fig.1) :

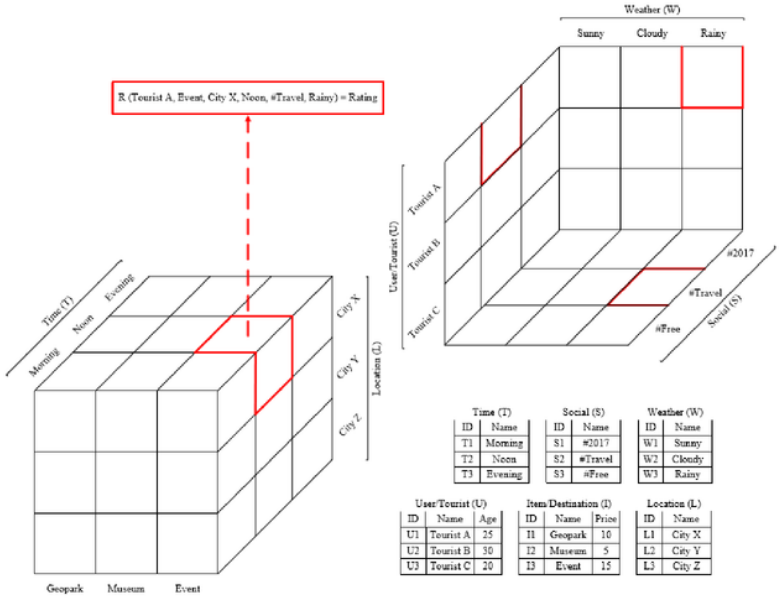

Fig.1: CARS Multidimensional model for tourism [2].

2.3.2 Choosing the architecture

Arguably, obtaining the information which will be used in the recommender system is a determining factor, especially in tourism and hospitality due to heterogeneity of this domain. [7] Although contextual information assists with overcoming certain drawbacks, Matrix and Tensor Factorization techniques are considered to be a promising solution to overcome further challenges related to data sparsity and scalability. [1]

Traditional Matrix Factorisation model is 2-dimensional whereas Tensor Factorization results in multi-dimensional matrix which is factored into lower-dimensional representation with user item and each contextual dimension representing a lower dimensional feature vector. [1] Alternatively, context is incorporated by using classical two-dimensional recommendation (user * item) methods. When applying the pre-filtering method context is used to filter out irrelevant ratings before they are used for computing recommendations, in post-filtering contextual information is used after the classical two- dimensional recommendation methods are applied to the non-contextual recommendation data and contextual modelling incorporates contextual information directly into its recommendation process. [1] In practise, one of the latter three are most commonly used.

2.3.3 Evaluation

Widely adopted measures for RS evaluation include mean absolute error, coverage, precision, recall, mean squared error and some derivatives of these. [8] On the other hand, considering how complex the product of tourism and hospitality is, alternative methods which are gaining more attention recently could be considered: topic diversification, coverage serendipity and means to measure novelty and diversity of the recommender results. [5] It also important to note that Rank Accuracy is an important metric as recommendation websites such as Yelp, TripAdvisor, Square Meal are structured to present products and services to the users in a certain order.

3.0 Problem Statement

Researchers and practitioners have identified various Recommender Systems limitations. The aim of this project is to further narrow down on the issues RS face in hospitality field and adopt a strong user-oriented approach.

The motivation behind this tactic is the lack of user confidence in how genuine TripAdvisor recommendations are [22], [23] – it is believed that recommendations are influenced by the business owners and are an ingenuine marketing tool to promote more affluent restaurants (or other services/attractions) which small independent establishments cannot compete with.

In addition, although information overload could be seen as a challenge, rapid technology improvements strive to overcome it, turn into positive but at an expense of 'data greed' [20] where user personal privacy may be compromised, as proved in recent events. [10]

The aim is to develop a restaurant recommender system which tackles the following limitations:

- Business owners influence over restaurant rating
- Personal privacy
- Explainability
- Scalability, sparsity and cold-start

The data will be gained by scraping restaurant reviews and basic user information from TripAdvisor. This means that no private user data will be used in this research and no offline marketing activities will influence the architecture of the system. Moreover, a randomised approach for data collection will allow all restaurants (part of the chain, independent, low/high rated and ranked) constitute the data set.

RS will be built using contextual collaborating filtering approach where reviews will be analysed using sentiment analysis to gain context, as oppose to taking usual approach of using users' social network and other personal data.

Evaluation techniques will involve estimating prediction accuracy on restaurant ratings with the help of context incorporation, where contextual features will be extracted from the reviews. Following this step, additional offline methods will be adopted to explore RS performance in the context of diversity, coverage, serendipity and novelty.

4.0 Project pipeline

4.1 Data Collection – the data set

TripAdvisor is considered to be the most widely used review and recommendation engine in travel and hospitality industry and has been chosen to be the source for the data collection. Although there are several readily available data sets online, the purpose of this project is to use a raw real-world data and simulate a real-world RS solution.

As RS primarily requires information on the user and the item, basic information on the user and restaurant reviews will be collected. It is important to stress that this project is user-focused, personal data sensitivity is important and the aim is experiment on constructing a recommender system which does not compromise user privacy.

The following attributes are planned to construct the data set:

User: username, location, the number of restaurant reviews published on the site

Restaurant: name, address, review date, cuisine, total number of reviews, rating, ranking amongst all London restaurants

In addition, a number of secondary features are planned to be added to the data set to constitute basis for contextual analysis:

Day of the week of the review, review length, user labelling (Londoner/Tourist) and potentially more, depending on the outcome of evaluation of the data set.

4.2 Data Collection – technique

Python is the chosen programming language for the project. Data collection will be carried out using Selenium library, which essentially automates the web browser interactions. The interaction in this instance will consist of scraping TripAdvisor's website and collecting the data outlined above.

There are a few web scraping networks available, however, a spider will be built from scratch following the preliminary logic outlined below: (Fig.2)

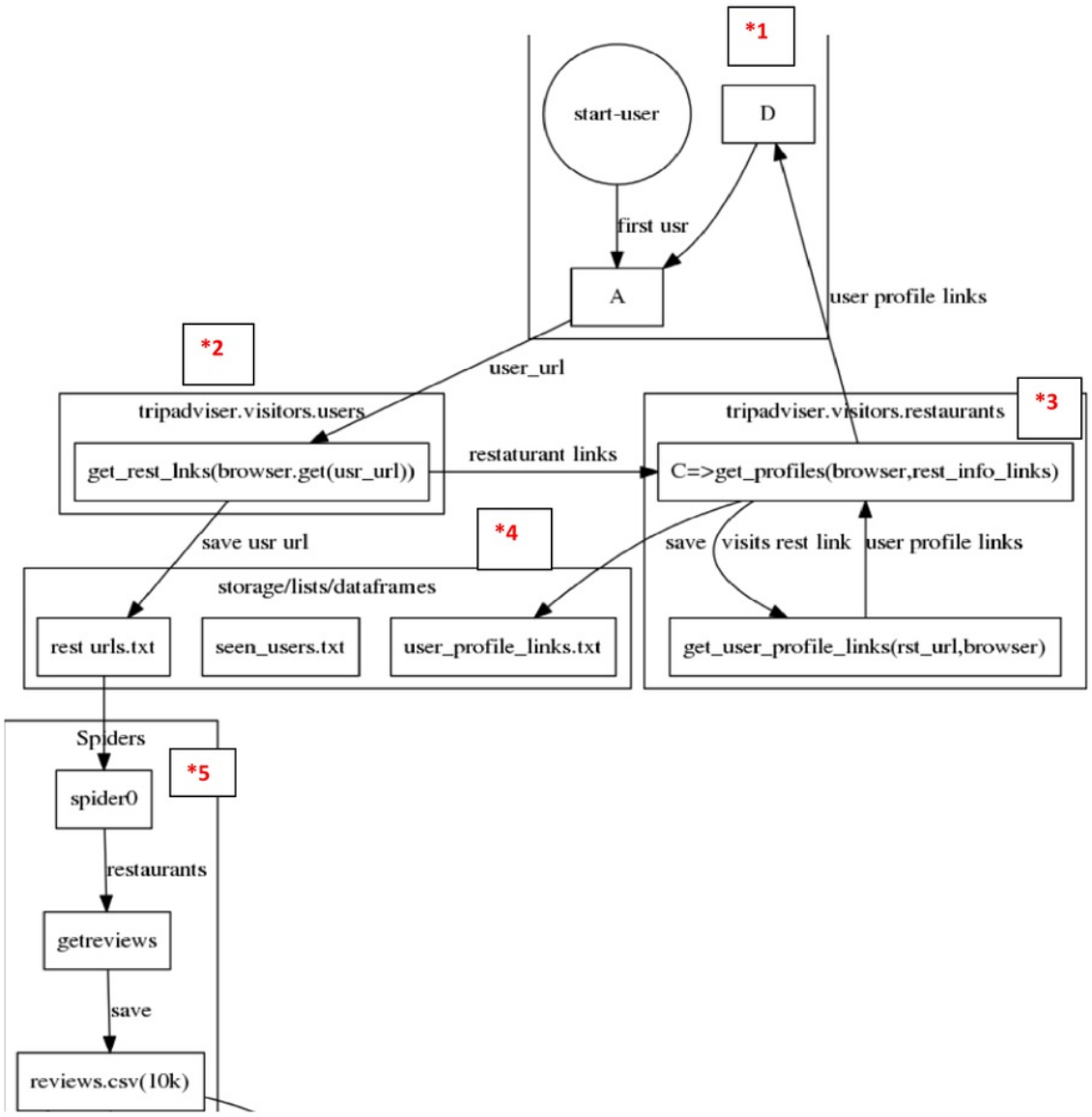

Figure 2: Preliminary logic of the spider

There will be elements of randomness in order to collect the data which consists of various London restaurants and collects reviews with a wide spectrum of ratings, restaurants from varied London locations left by users from various locations. This idea supports the intention of creating an RS which is independent of any business-driven influences current TripAdvisor RS may have.

Starting from a random user profile it will collect user information and collect the links of restaurants the user has reviewed. *1, *2

It will continue onto exploring those restaurants, collecting user information from each of the restaurant links and storing all the information outlined above long the way. *3, *4

The spider will be controlled to check for duplicates and will contain limiting features to control the amount of data collected. The intention is to collect up to 10000 reviews and report on any overlaps (for example, where a user has reviewed multiple restaurants). *5

4.3 Contextual opinion extraction

Contextual features will be mined using Python's SpaCy library – an industrial-strength natural language processing library. Following the direction of previous research [13] reviews will be classified into positive, neutral and negative, based on the adjectives used. This aspect will support and influence the rating predictions as evaluation steps will be taken to compare the true rating and rating proposed by the context of the review.

Further, relevant reoccurring topics will be extracted from the reviews. Previous experiments [27] carried out analysis where contextual topics included: food, ambience, service, deals and worthiness. The topics will not be outlined prior to starting the experiments as raw data will be used, and the aim is to simulate a real-life problem where decisions will be based on the evaluation of the data.

Finally, the attributes will be assigned contribution weights (or 'importance scores') using 'Term Frequency, Inverse Document Frequency' which is part of the SpaCy library referenced earlier in this proposal. Another potential idea (depending on the outcome of sentiment analysis) is to associate the features with either user-aligned or item-aligned scenario and apply Support Vector Machines to filter and identify the most relevant items to be incorporated in the next steps of building RS architecture.

4.4 Building the Recommender System

The purpose of this project is to experiment with building an innovative recommender system that overcomes the drawbacks identified by RS researchers and practitioners as well as addressing user concerns over personal data privacy as well as authenticity and quality of the recommendations.

To begin with, the recommender system will be built using context aware collaborative filtering approach and employ algorithms estimating rating accuracy. Incorporation of context extracted from user reviews addresses cold start and sparsity issues as well as tackles personal data privacy concerns.

The data set will be divided 60/20/20 into training, validation and testing sets respectively.

4.4.1 Matrix factorisation

Research has summarised that tensor factorisation algorithms are computationally expensive especially with large data sets. [15], [6] Non-negative matrix factorisation will be the first algorithm of choice. Although matrix factorisation is 2-dimensional by nature, we will integrate contextual features using all 3 approaches (pre filtering, postfiltering and contextual modelling) referenced above to overcome multidimensionality limitations.

Matrix factorization algorithm will form the user feature matrix U and the restaurant feature matrix I from the rating matrix R. Recommender algorithms regard R as a matrix with missing values, estimate U and I based on known values, and then the missing values are estimated by U*I. [15]

4.4.2 Similarity measures

For the purpose of experimentation and comparison, we will also adopt more traditional collaborative filtering algorithms based on predictions which are estimated using similarity measures. Namely, we will experiment with 2 approaches previously adopted in research which experimented with RS based on user reviews. [13]

CF: memory-based. Here we will run KNN algorithm twice: in case of user-based and item-based scenario. The algorithm will identify k users (used based scenario) and k restaurant (items -based scenario) most similar to each other with the help of Pearson correlation coefficient. Ratings for the 'unrated' restaurants will be predicted based on similarity results.

Based on the predicted ratings of the restaurants we will aim to analyse the recommendations presented by the system. Moreover, we will take an additional step and experiment with ranking algorithms which, as oppose to rating prediction, will obtain a list top-n restaurant recommendations. This will be the basis for exploring beyond-accuracy measures in the evaluation stage.

4.4.3 Bayesian Personalised Ranking (BPR)

This approach will also adopt matrix factorisation and has been previously used in RS research where building the system with minimal personal data requirement was one of the main goals. [20] Moreover, it has been recognised as the most popular ranking approach in modern RS. Our algorithm will look at a pair of restaurants for a particular user (one rated and one unrated) and with the help of stochastic gradient descent throughout the learning stage it will assign the scores to the restaurants and produce the ranking according to that score.

4.4.4 Beyond-accuracy measures

Previous research [18] has explored beyond – similarity measures conducting offline experiments with MovieLens and Last.fm datasets and suggested that future research should focus on RS in different domains. Restaurant recommender system built on a raw data (in our case) forms an unexplored real-life problem.

Adopting authors' approach [18] we will re-rank the list of restaurants produced by state-of the-art BPR, matrix factorisation and KNN algorithms used above 3 times: incorporating separately calculated novelty, diversity and surprise scores based on restaurant and user items in our data set.

4.5 Evaluating the results

Following the standard practise in RS research we will start evaluating our recommendation accuracy using the precision and recall metrics. Moreover, we will explore how rating and ranking accuracy results produced by matrix factorisation, KNN and BPR differ from each other.

It is apparent that re-ranking results would be best evaluated with user involvement [18] however, our offline experiment will measure how ranking results change for each algorithm with incorporation of each beyond-accuracy measure.

Eventually we will compare our results with the ratings and the rankings of restaurants produced by TripAdvisor. Users have recognised the portal is somewhat ingenuine due to suspected influence of restaurant owners and lack of knowledge on how the recommendations are produced. Our results will be transparent in terms of methods used and produced with genuine accessible data that does not compromise user privacy.

5.0 Conclusion

We have outlined a proposal for a project to build a Recommender System for London restaurants, tackling the drawbacks recognised in previous work in the RS domain. Moreover, we have built a plan to analyse the results using beyond-accuracy measures which previously have been mainly studied outside the field of hospitality. The Recommender System could then be refined for usage on a wider geographical area as well as other service and entertainment establishments.

The summary of the project pipeline.

		Date															
Step	Task	June				July				August				September			
		Week 1	*Week 2*	*Week 3*	*Week 4*	*Week 1*	*Week 2*	*Week 3*	*Week 4*	*Week 1*	*Week 2*	*Week 3*	*Week 4*	*Week 1*	*Week 2*	*Week 3*	*Week 4*
1	Build the spider which will scrape the data from TripAdvisor website																
2	Conduct exploratory data analysis and build the data set with secondary features																
3	Perform sentiment analysis and topic modelling on the reviews, prepare the features for the incorporation into the main prediction algorithms (weights and/or SVM)																
4	Initial implementatation of matrix factorisation, KNN, BRP on a small sample of the data set and exploring the results																
5	Apply KNN, BRP and matrix factorisation on the complete data set and incorporate the contextual information into the prediction process																
6	Estimate novelty, diversity and surprise scores based on restaurant and user items in the data set																
7	Apply KNN, BRP and matrix factorisation on the complete data set and incorporate each of the beyond-accuracy measures																
8	Analyse the RS results discuss how it compares to TripAdvisor rating and ranking																
9	Produce the Project Report																

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